

INSTALLATION & SETUP GUIDE

Arena Station

Setting up a new station on a Raspberry Pi

A step-by-step handbook for installing, wiring, and configuring a new Arena game station — written so your on-site team can do it without developer assistance.

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Applies to: Raspberry Pi 4 / Pi 5 stations

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1 Overview — how a station works

Each station is a small Raspberry Pi computer connected to a card reader, two buttons, and a set of relays. It talks to the central Arena backend server over your local network.

In normal play the flow is simple:

1. A player **scans their RFID band** on the reader → the game session starts automatically.
2. The timer runs and the game becomes active.
3. The player can press the **Hint button** to request a hint.
4. The player presses the **Stop button** (or staff scans a card) → the session ends and results are sent to the backend.

Two services run on every station

`arena-station` is the main application (reads cards/buttons, drives relays, talks to the backend). `arena-watchdog` is a tiny helper that lets your admin PC restart a station remotely if it ever hangs. The installer sets up **both** automatically.

You install everything by running **one command** on each new Raspberry Pi. The rest of this guide walks through wiring the hardware first, then running that command, then confirming the station is online.

2 What you'll need

Hardware (per station)

- ☐ Raspberry Pi 4 or Pi 5 with power supply and a microSD card (16 GB+)
- ☐ PN532 NFC/RFID reader (I2C model)
- ☐ 2 × momentary push buttons (Stop and Hint)
- ☐ 1 × 3-channel relay module (active-low)
- ☐ Jumper wires
- ☐ RFID cards / wristbands for players
- ☐ Network connection — Ethernet cable (recommended) or Wi-Fi

Information to have ready

- ☐ The **Backend server address** (e.g. `http://192.168.50.10:8000`)
- ☐ A unique **Station ID** for this unit (e.g. `station-3`)
- ☐ A friendly **Station name** (e.g. `Laser Room`)
- ☐ The **Watchdog token** — must match the value configured on the backend

Tip — get the network right first

Put every station and the backend server on the **same local network**

. Wired Ethernet is far more reliable than Wi-Fi for a fixed installation. It also helps to reserve a fixed IP address for each Pi in your router.

3 Wiring the hardware

Power the Raspberry Pi off before wiring. All pin numbers below use the physical **board pin** position where noted, and the **BCM/GPIO number** that the software expects.

3.1 PN532 NFC / RFID reader (I2C mode)

PN532 pin	Connect to Raspberry Pi
VCC	3.3V — physical Pin 1
GND	GND — physical Pin 6
SDA	GPIO 2 / SDA — physical Pin 3
SCL	GPIO 3 / SCL — physical Pin 5

Set the PN532 to I2C mode

On the small DIP switch on the reader:

Switch 1 = OFF

,

Switch 2 = ON

. If the reader is in UART or SPI mode it will not be detected.

3.2 Buttons

Button	GPIO pin	Wiring
Stop button	GPIO 17	One leg to GPIO 17, the other leg to any GND
Hint button	GPIO 27	One leg to GPIO 27, the other leg to any GND

No external resistors are needed — the internal pull-ups are enabled in software.

3.3 Relay module (3-channel, active-low)

Relay input	GPIO pin	Purpose
IN1	GPIO 5 (physical Pin 29)	Game Active indicator
IN2	GPIO 6 (physical Pin 31)	Reset / Stop indicator
IN3	GPIO 13 (physical Pin 33)	Ready / Idle indicator
VCC	5V (physical Pin 2)	Relay power
GND	GND (physical Pin 9)	Ground

These pin numbers are the defaults

They match the software's out-of-the-box settings, so if you wire as shown above you won't need to change anything. If your build uses different pins, they can be adjusted later in the configuration file (see Section 8).

4 Preparing the Raspberry Pi

1 Flash Raspberry Pi OS

Use the official **Raspberry Pi Imager** to write **Raspberry Pi OS (64-bit)** to the microSD card. In the Imager's advanced settings you can pre-set the hostname, enable SSH, and enter your Wi-Fi details to save time.

2 Boot and connect to the network

Insert the card, connect the Ethernet cable (or confirm Wi-Fi), and power on the Pi. Confirm it can reach the internet — the installer downloads software packages.

3 Enable I2C (needed for the card reader)

Open a terminal on the Pi and run:

```
sudo raspi-config
```

Go to **Interface Options** → **I2C** → **Enable** → **Yes**, then finish and reboot if asked. After reboot you can confirm the reader is seen:

```
sudo i2cdetect -y 1
```

A device at address `0x24` (or `0x48`) means the NFC reader is wired correctly.

You only do Section 4 once per Pi

Once the OS is flashed and I2C is on, everything else is handled by the automated installer in the next section.

5 Installing the station software

This is the main step. One command downloads the installer and sets up everything: system packages, the application, configuration, and both services.

On the Raspberry Pi, open a terminal and run these two lines:

```
curl -fsSL https://raw.githubusercontent.com/m32jawad/Arena/main/station/setup_station.sh -o
setup_station.sh

bash setup_station.sh
```

The installer will automatically:

- Install the required system packages (Git, GPIO and I2C libraries).
- Download the station application.
- Install the Python dependencies.
- Ask you a short list of configuration questions (see Section 6).
- Install and start the `arena-station` and `arena-watchdog` services so the station starts automatically every time the Pi is powered on.

Safe to re-run any time

Running the command again simply updates the software and re-applies your settings. Your previous configuration file is backed up first, so nothing is lost.

If you are asked for a password

The installer needs administrator (sudo) rights to install system components. Enter the Pi user's password when prompted — this is normal.

6 Answering the setup questions

During installation you'll be asked the following. Press **Enter** to accept the default shown in brackets, or type your own value.

Question	What to enter	Example
Station ID	A unique short ID for this station. Must be different on every station.	station-3
Station display name	A friendly name shown on screens.	Laser Room
Backend URL	Address of your central Arena server, including the port.	http://192.168.50.10:8000
Station server port	Port the station listens on. Leave as default unless told otherwise.	8001
Watchdog token	Shared secret for remote restart. Must match the value set on the backend.	arena-station-secret
Watchdog port	Port for the remote-restart helper. Leave as default.	8002
Enable NFC/RFID reader?	Answer y — unless this station has no card reader.	y
Simulate hardware?	Answer n on a real station. (y is only for testing without any hardware.)	n
Enable debug logging?	Answer n for normal use.	n

Two values matter most

Backend URL

must point at your real server, and the

Watchdog token

must be identical to the one on the backend. If either is wrong, the station won't connect or can't be restarted remotely.

7 Verifying the station works

When the installer finishes it prints "**Setup complete**". Confirm the station is healthy:

1 Check the service is running

```
sudo systemctl status arena-station.service
```

You should see **active (running)** in green.

2 Check the station responds

```
curl http://localhost:8001/
```

This returns a short status message showing the station ID, name and backend URL.

3 Test a real card and buttons

Scan an RFID band on the reader — a session should start. Press the Hint and Stop buttons and confirm the game responds. The station should now also appear as **online** on your backend/admin dashboard.

It starts on its own

Because both services are enabled, the station comes back up automatically after a power cut or reboot — no need to log in and start anything.

8 Everyday commands

Handy commands for day-to-day operation. Run them in a terminal on the Pi.

Task	Command
Check if the station is running	<code>sudo systemctl status arena-station.service</code>
Watch live logs	<code>sudo journalctl -u arena-station.service -f</code>
Restart the station	<code>sudo systemctl restart arena-station.service</code>
Stop the station	<code>sudo systemctl stop arena-station.service</code>
Edit the configuration	<code>nano ~/arena/station/.env</code>

Changing a setting later

All settings live in the file `~/arena/station/.env`. To change one — for example the backend address or a GPIO pin — edit the file and then restart the station so the change takes effect:

```
nano ~/arena/station/.env
sudo systemctl restart arena-station.service
```

Remote restart

The `arena-watchdog` service lets your admin PC restart this station over the network (using the Watchdog token) without anyone visiting the machine. It runs on port `8002` and starts automatically.

9 Troubleshooting

The card reader isn't detected

- Confirm I2C is enabled: `sudo raspi-config` → Interface Options → I2C.
- Check wiring, then run `sudo i2cdetect -y 1` — you should see `0x24` (or `0x48`).
- Confirm the PN532's DIP switch is in I2C mode (Switch 1 OFF, Switch 2 ON).

The station can't reach the backend

- Verify the **Backend URL** in `~/arena/station/.env` is correct.
- From the Pi, test the server: `curl http://192.168.50.10:8000/api/auth/public/storylines/` (use your real server address).
- Make sure the Pi and the backend are on the same network and the server's firewall allows port `8000`.

Buttons or relays don't respond

- Re-check the GPIO wiring against Section 3.
- View the logs while pressing a button: `sudo journalctl -u arena-station.service -f`.
- Confirm **Simulate hardware** was set to **n** during setup.

The service won't start

- Read the error in the logs: `sudo journalctl -u arena-station.service -n 50`.
- Re-run the installer — it repairs the install and is safe to run again.

Still stuck?

Copy the last 50 log lines (`sudo journalctl -u arena-station.service -n 50`) and send them, along with the Station ID, to your technical contact. That is almost always enough to diagnose the problem.

10 Information to record for each station

Keep a simple record for every station you install. It makes support and future changes much easier.

Field	Value for this station
Station ID	
Station name	
Physical location	
Raspberry Pi IP address	
Backend URL used	
Date installed	
Installed by	

Quick reference — the whole install in short

1. Flash Raspberry Pi OS, connect to network, enable I2C.
2. Wire the reader, buttons and relays (Section 3).
3. Run the two installer commands (Section 5).
4. Answer the setup questions (Section 6).
5. Verify with `systemctl status` and a real card scan (Section 7).